LOTA-MP: Multi-Patch Evidence Aggregation for Bit-Plane Guided AI-Generated Image Detection

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GitHub: `https://github.com/SundaramPandey2005/LOTA-AI-Image-Forensics`

I. Abstract (Maximum 100 Words)

1. Reconstruction-error detectors (DIRE, LaRE^2) need a diffusion model and multi-step sampling for every image, and generalise poorly beyond the generator family they were trained on.
2. They measure what a generator can re-synthesise, a generator-specific cue costing 0.26–2 s per image.
3. LOTA (ICCV 2025) uses the three lowest bit-planes, which hold only imperceptible noise: it normalises them into a noise image, keeps the 32×32 patch with maximal multi-directional gradient, and classifies it with ResNet-50.
4. Detection becomes a millisecond, generator-agnostic test of sensor-level statistics instead of image semantics.

II. Proposed Novelty & Technical Contribution

1. **LOTA-MP (Multi-Patch evidence aggregation).** LOTA decides from *one* 32×32 patch, about 1.5% of the image, and its own ablation (Table 5 of the paper) shows that the choice of patch moves BigGAN accuracy from 94.9% to 69.9%. We instead keep the K highest-gradient patches, score each with the same network, and fuse their log-odds into one image-level decision.

2. Baseline: $\hat{y} = \sigma(f_\theta(\tilde{z}_{p^*}))$, $p^* = \arg\max_p g_p$. Ours: $\hat{y} = \sigma\big(\frac{1}{K}\sum_{k=1}^{K} f_\theta(\tilde{z}_{(k)})\big)$ over the K patches ranked by g_p, with the loss on the fused logit so training matches inference. Patch selection changes from arg-max to top-K and a parameter-free fusion is added; no layer or weight is added, K=1 recovers LOTA exactly, we use K=4.

III. Theoretical / Mathematical Justification

1. Let $s_k = f_\theta(\tilde{z}_{(k)})$ be the logit of the k-th ranked low-bit patch and $y \in \{0, 1\}$ the label (1 = generated). LOTA-MP uses $\bar{s} = \frac{1}{K}\sum_{k=1}^{K} s_k, \quad \mathcal{L} = -\,y \log \sigma(\bar{s}) - (1-y)\log\big(1-\sigma(\bar{s})\big)$.

2. *Why the mean of logits.* For patches that are conditionally independent given y and balanced classes, Bayes' rule gives $\log \frac{p(y=1|\tilde{z}_{(1:K)})}{p(y=0|\tilde{z}_{(1:K)})} = \sum_k \log \frac{p(\tilde{z}_{(k)}|1)}{p(\tilde{z}_{(k)}|0)}$: the optimal statistic is a *sum of per-patch log-odds*, whose sign equals that of $\bar{s}$. *Why it helps.* Write $s_k = \pm\mu + \varepsilon_k$ (+ for generated), with $\mathrm{Var}[\varepsilon_k] = \sigma^2$ and pairwise correlation ρ. Then $\mathbb{E}[\bar{s}] = \pm\mu$ and $\mathrm{Var}[\bar{s}] = \sigma^2\big(\rho + \frac{1-\rho}{K}\big)$, so for Gaussian ε the error is $P_e(K) = Q\Big(\frac{\mu}{\sigma}\sqrt{\frac{K}{1+(K-1)\rho}}\Big) \le Q\big(\frac{\mu}{\sigma}\big) = P_e(1)$, strictly smaller whenever $\rho < 1$. In training, $\partial\mathcal{L}/\partial s_k = (\sigma(\bar{s}) - y)/K$: every patch receives gradient, so each image supplies K patch samples per step, and since $\mathcal{L}$ is convex in the logit, $\mathcal{L}(\bar{s}) \le \frac{1}{K}\sum_k \mathcal{L}(s_k)$ (Jensen), so uninformative patches are not forced to be individually confident.

3. The gain depends on the single-patch margin μ/σ and the inter-patch correlation ρ: largest where one patch is unreliable (unseen generators), bounded by $Q(\mu/(\sigma\sqrt{\rho}))$ as $K \to \infty$ (hence a small K), zero where the baseline saturates. The price is K backbone passes per image.

IV. Comparative Results & Empirical Evaluation

1. Tiny-GenImage, a public subset of the paper's GenImage benchmark: 7 generators (BigGAN, Midjourney, Wukong, SD v1.5, ADM, GLIDE, VQDM), 2,000 + 2,000 training and 500 + 500 test images each. Both models are trained on SD v1.5 (10% held out for checkpoint selection) and tested on all 7 generators at original resolution; ResNet-50, Adam, lr 10^{-4}, 10 epochs, 3 seeds, identical data, splits, seeds and schedule, only K differs.

2. Table 1: accuracy (threshold 0.5) and average precision, averaged over the 7 test generators; mean ± std over seeds. Latency is per image at batch size 1 on a Tesla T4 (Kaggle).

3. LOTA-MP raises average accuracy by +15.1 points (per seed +16.6, +16.8, +12.0) and AP by +15.7 points with identical parameters and latency. Trained on BigGAN instead: 62.3 → 70.0 (+7.7).

Table 1: Quantitative comparison of the baseline and proposed method on Tiny-GenImage (trained on SD v1.5, tested on 7 generators). Best results are highlighted in bold.

Method	Avg. ACC (% ↑)	Avg. AP (% ↑)	Params (M)	Latency (ms ↓)
Baseline: LOTA (reproduced, K=1)	67.1 ± 1.7	76.0 ± 1.5	23.5	10.8
LOTA + multi-patch fusion at inference only (K=4)	74.0 ± 3.3	84.3 ± 2.3	23.5	10.7
Ours: LOTA-MP (K=4)	**82.2 ± 1.1**	**91.7 ± 0.8**	23.5	10.7
Δ (Ours – Baseline)	+15.1	+15.7	+0.0	−0.1

Result Interpretation: The gain holds on every generator and seed and is largest on unseen generators (Midjourney +19.4, ADM +14.6, VQDM +14.1), as Sec. III predicts for small single-patch margins. Test-time fusion alone gives +6.9 (row 2); training on the fused logit adds +8.2 and cuts seed variance (±3.3 → ±1.1). Accuracy rises monotonically with K (72.4/77.6/82.2/85.1 for K=1/2/4/8) with diminishing returns, as the $(1-\rho)/K$ term predicts; the log-odds mean beats probability averaging (80.7) and voting (76.7). *Trade-offs.* Latency is unchanged because the K patches are batched through one backbone pass (K=8: 13.6 ms), but training is 3× slower (10.6 vs. 3.6 min). Under unseen blur (σ=1) or JPEG-90 both models drop to ≈ 50%: the degradation erases the low-bit cue itself, which no aggregation can recover. *Caveat.* Our baseline (67.1%) is far below the paper's 98.9% because Tiny-GenImage has 4k training images per generator instead of ∼160k and we train 10 epochs (both models were still improving). The absolute numbers are a low-data regime; the comparison is controlled, and this regime is where K patch samples per image should matter most.